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Uniprocessor Scheduling

单处理机调度



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Uniprocessor Scheduling

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Type of processor scheduling 处理机调度的类型

- Long-term scheduling
- Medium-term scheduling
- Short-term scheduling

长/中/短

Scheduling Algorithms 调度算法

- Short-term scheduling criteria 短期调度准则
- The use of priorities 优先级的使用
- Alternative scheduling policies 备选调度策略
- Performance comparison 性能对比
- Fair-share scheduling 公平共享调度

Processor Scheduling

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多道程序系统的核心：调度

- Key of multiprogramming system: scheduling
- Aim to assign processes to be executed by the processor in a way that meets system objectives, such as response time, throughput, and processor efficiency

处理器利用率　响应时间　吞吐量

- Four types of scheduling are typically involved:

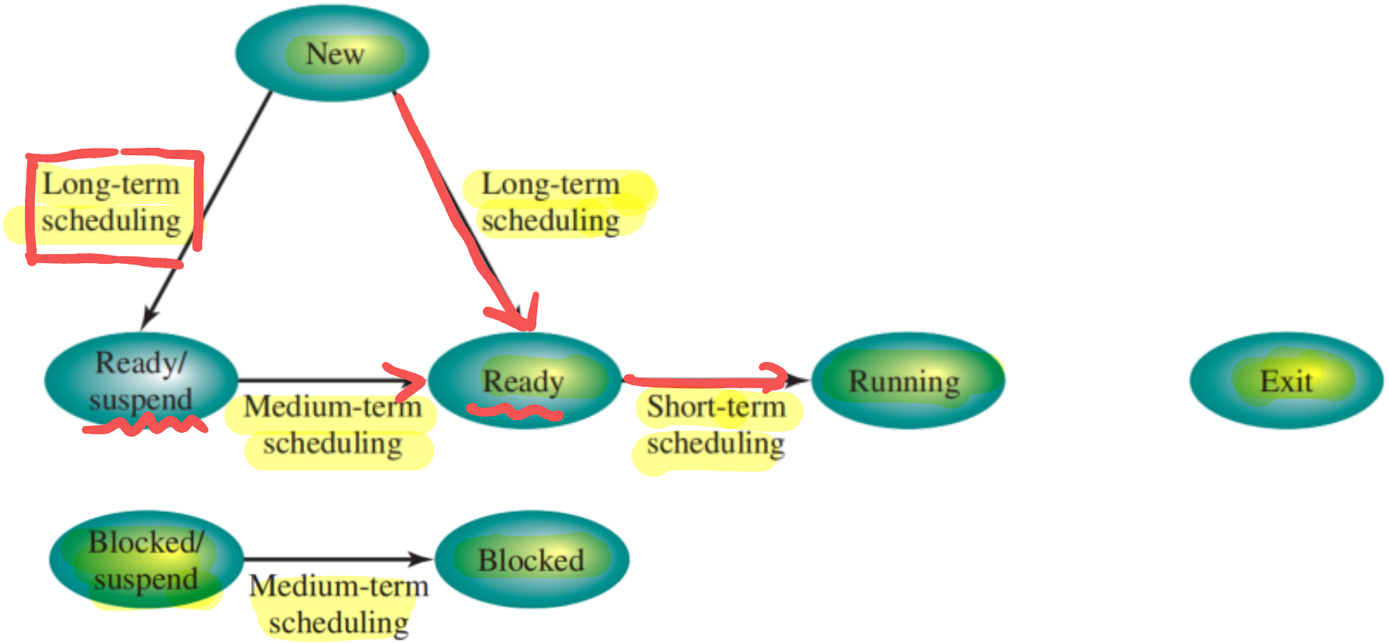
Long-term scheduling	The decision to add to the pool of processes to be executed.
Medium-term scheduling	The decision to add to the number of processes that are partially or fully in main memory.
Short-term scheduling	The decision as to which available process will be executed by the processor.
I/O scheduling	The decision as to which process's pending I/O request shall be handled by an available I/O device.

下一步　部分/全部　挂起　处理

Scheduling and Process State Transitions

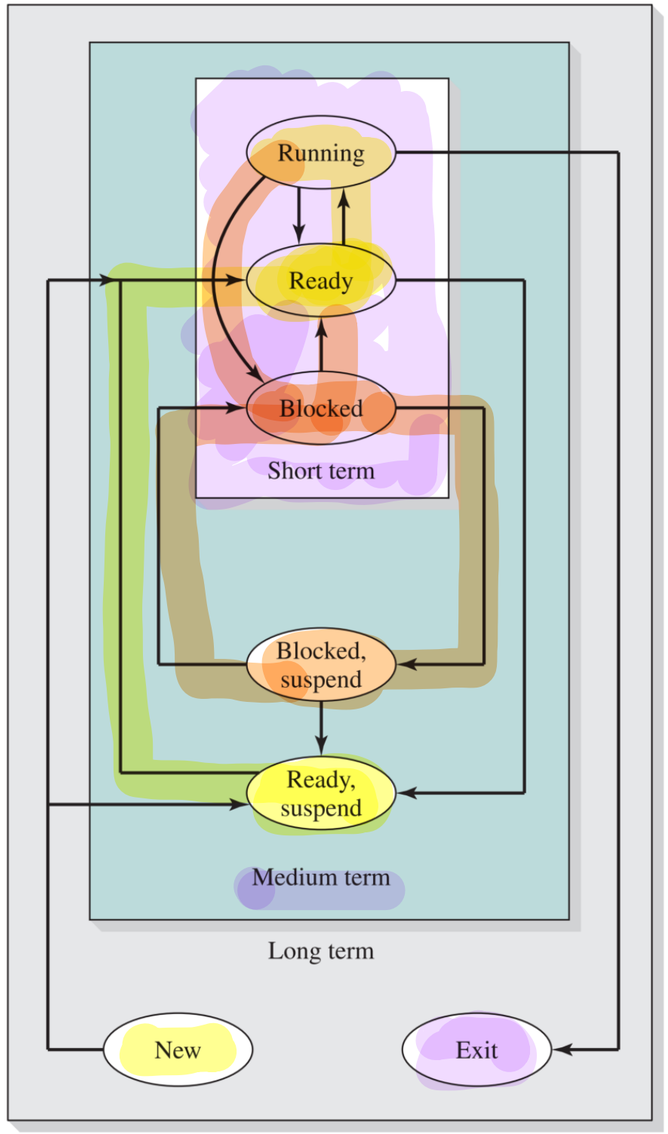
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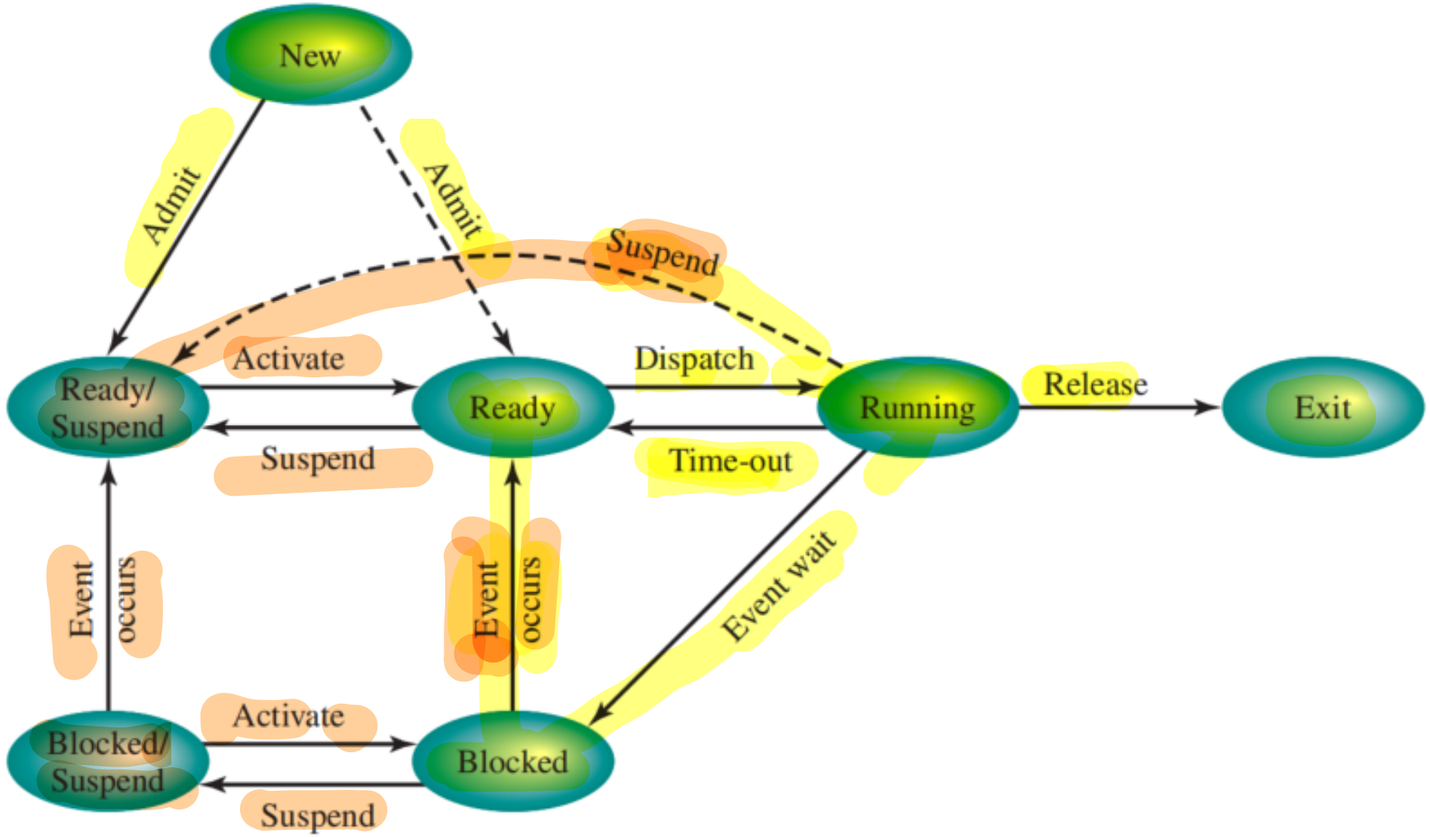
- Long-term scheduling is performed when a new process is created.
- Medium-term scheduling is a part of the swapping function.
- Short-term scheduling is the actual decision of which ready process to execute next.



Nesting of Scheduling Functions

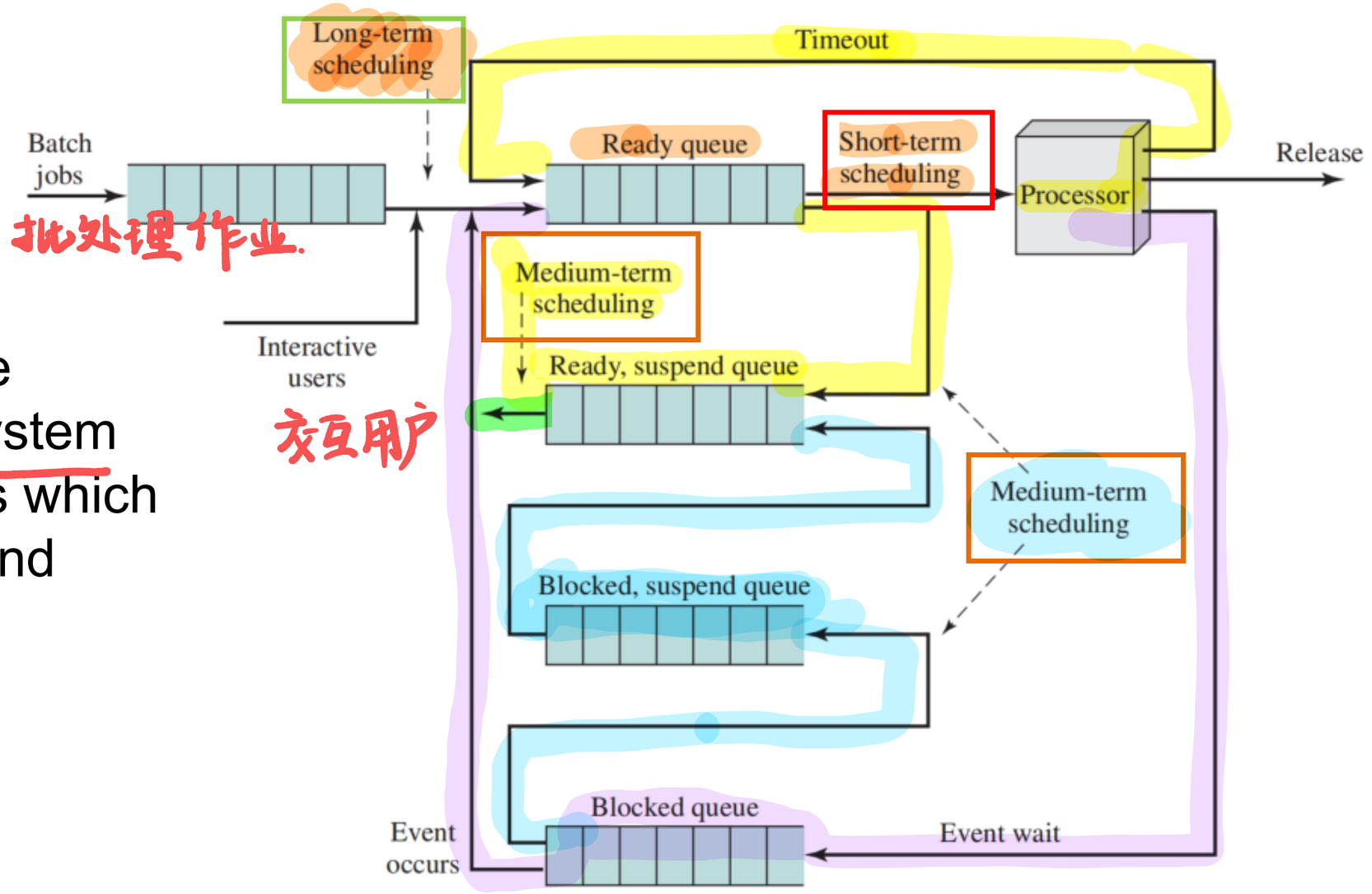
功能调度的嵌套关系





Queuing Diagram

排队图



- Scheduling affects the performance of the system because it determines which processes will **wait**, and which will **progress**

Long-Term Scheduling

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- **Determines which programs are admitted to the system for processing**
- **Controls the degree of multiprogramming**
  - the more processes that are created, the smaller the percentage of time that each process can be executed
  - may limit to provide satisfactory service to the current set of processes
- **Creates processes from the queue when it can, but must decide:**
  - when the operating system can take on one or more additional processes
  - which jobs to accept and turn into processes
    - first come, first served
    - priority, expected execution time, I/O requirements

Medium-Term Scheduling

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- Part of the swapping function 内存反换功能
- Swapping-in decisions are based on the need to manage the degree of multiprogramming
  - considers the memory requirements of the swapped-out processes

基于对多道程序度的管理需求

内存需求

Short-Term Scheduling

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进程调度

- Known as the dispatcher
- Executes most frequently
- Makes the fine-grained decision of which process to execute next
- Invoked when an event occurs that may lead to the blocking of the current process or that may provide an opportunity to preempt a currently running process in favor of another

阻塞

触发

(抢占)

抢占当前运行进程以执行其他进程

  - Examples:
    - Clock interrupts
    - I/O interrupts
    - Operating system calls
    - Signals (e.g., semaphores)

信号量

Short Term Scheduling Criteria

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- **Main objective is to allocate processor time to optimize certain aspects of system behavior**
- **A set of criteria is needed to evaluate the scheduling policy**
  - user-oriented and system-oriented criteria
  - performance related and not directly performance related

Two dimensions

One Dimension

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User-oriented criteria

- relate to the behavior of the system as perceived by the individual user or process (such as **response time** in an interactive system)
- important on **virtually all systems**

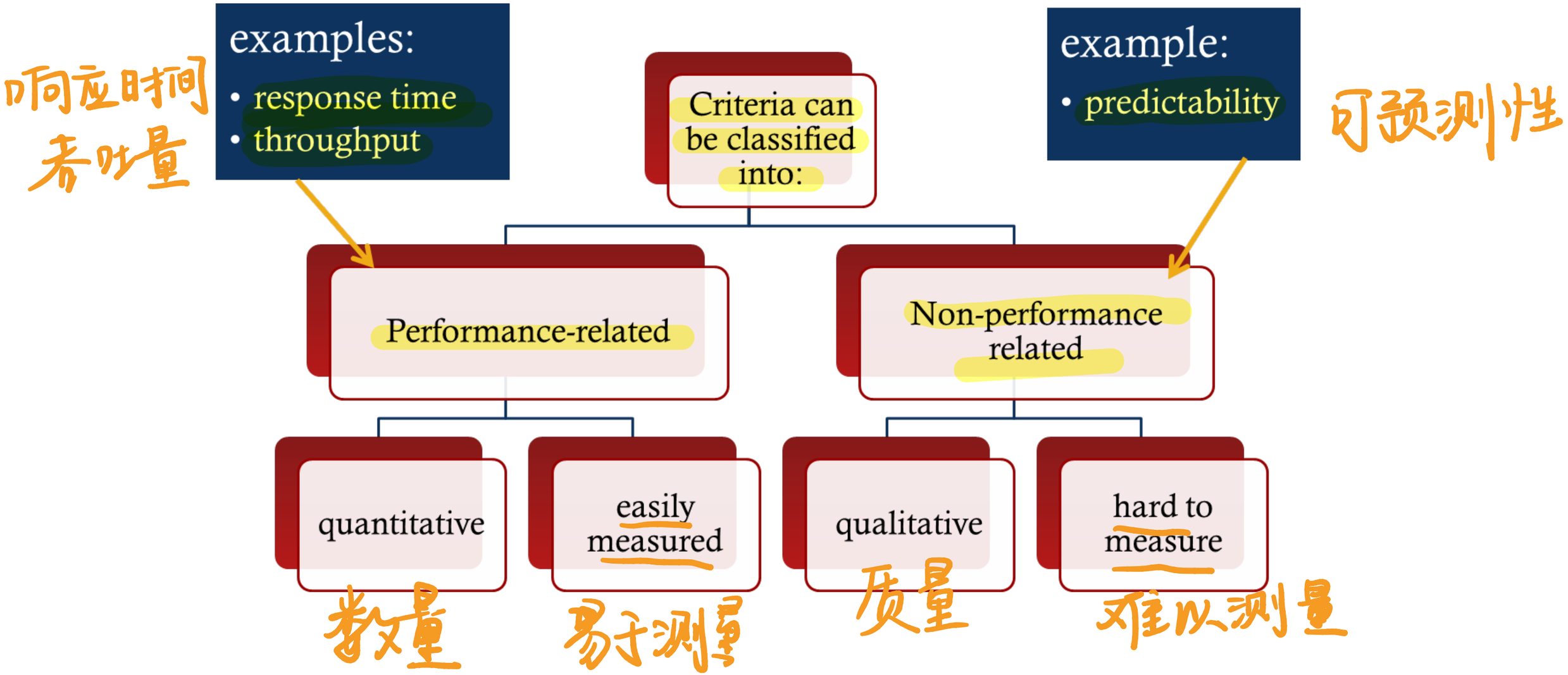
System-oriented criteria

- focus in on **effective and efficient** utilization of the processor (rate at which processes are completed)
- generally of **minor importance** on single-user systems

用户导向：与单个用户或进程感知的系统行为相关【例如：有交互式系统的响应时间】，几乎适用于所有系统

系统导向：高效利用，进程完成率，在单用户系统次要

Another dimension



Scheduling Criteria

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周转时间：
用户导向·性能相关
实际执行时间+等待时间

User Oriented, Performance Related

Turnaround time This is the interval of time between the submission of a process and its completion. Includes actual execution time plus time spent waiting for resources, including the processor. This is an appropriate measure for a batch job.

Response time For an interactive process, this is the time from the submission of a request until the response begins to be received. Often a process can begin producing some output to the user while continuing to process the request. Thus, this is a better measure than turnaround time from the user's point of view. The scheduling discipline should attempt to achieve low response time and to maximize the number of interactive users receiving acceptable response time. 响应时间：从提交请求到开始接收响应时间

Deadlines When process completion deadlines can be specified, the scheduling discipline should subordinate other goals to that of maximizing the percentage of deadlines met.

截止时间：最大完成截止时间达标率

User Oriented, Other

可预测性 无论系统负载如何，同一作业的运行时间和成本应基本稳定

Predictability A given job should run in about the same amount of time and at about the same cost regardless of the load on the system. A wide variation in response time or turnaround time is distracting to users. It may signal a wide swing in system workloads or the need for system tuning to cure instabilities.

吞吐量

System Oriented, Performance Related

提升单位时间内完成的进程数

Throughput The scheduling policy should attempt to maximize the number of processes completed per unit of time. This is a measure of how much work is being performed. This clearly depends on the average length of a process, but is also influenced by the scheduling policy, which may affect utilization.

Processor utilization This is the percentage of time that the processor is busy. For an expensive shared system, this is a significant criterion. In single-user systems and in some other systems, such as real-time systems, this criterion is less important than some of the others.

处理器占用率：处于忙碌时间占比

System Oriented, Other

公平性：

Fairness In the absence of guidance from the user or other system-supplied guidance, processes should be treated the same, and no process should suffer starvation.

Enforcing priorities When processes are assigned priorities, the scheduling policy should favor higher-priority processes.

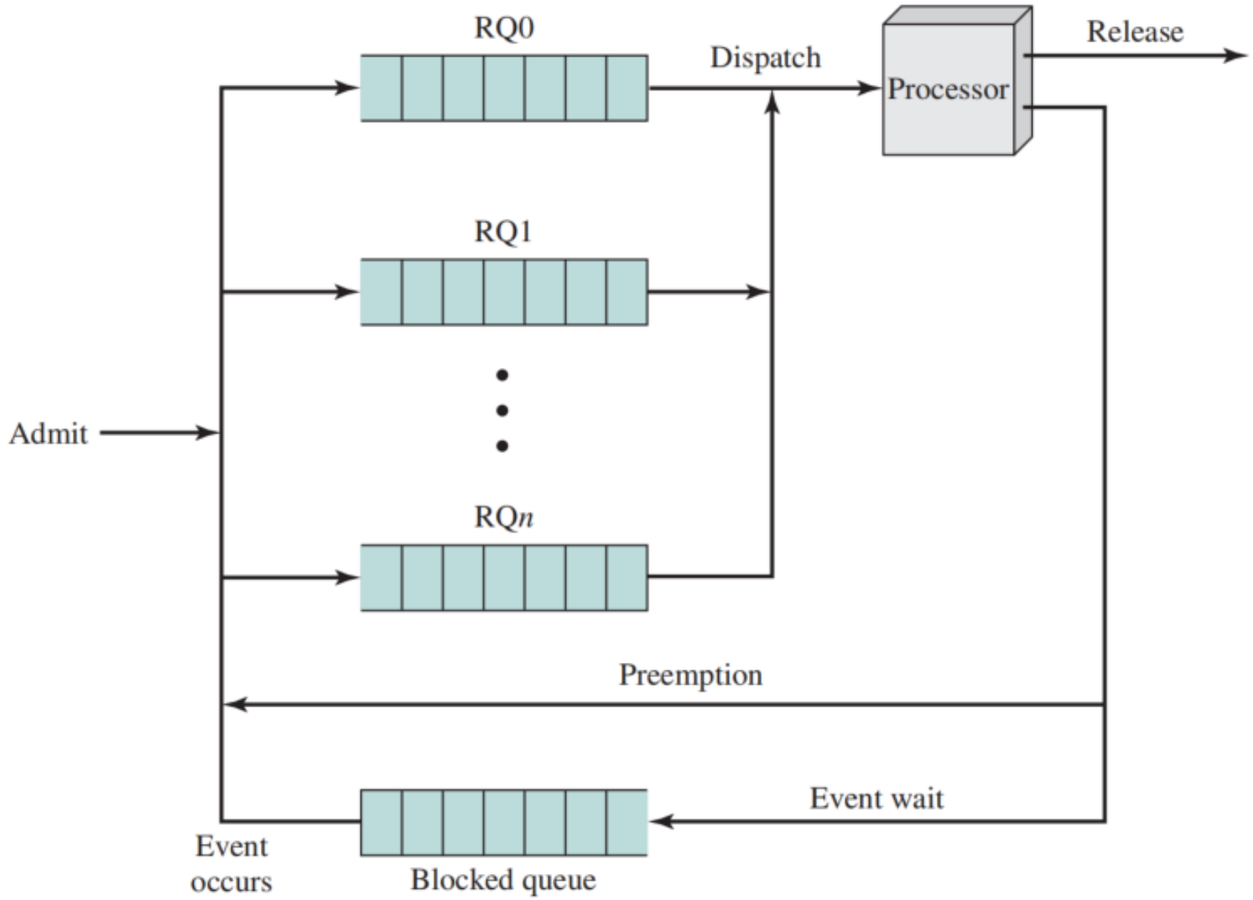
优先级强制执行

Balancing resources The scheduling policy should keep the resources of the system busy. Processes that will underutilize stressed resources should be favored. This criterion also involves medium-term and long-term scheduling.

The Use of Priorities

- In many systems, each process is assigned a priority, and the scheduler will always choose a process of higher priority over one of lower priority.
- If there are one or more processes in the queue, a process is selected using some scheduling policy. If RQ0 is empty, then RQ1 is examined, and so on.
- Pure priority scheduling scheme is that lower-priority processes may suffer starvation

larger-priority values represent lower-priority processes



Priority Queuing

Alternative Scheduling Policies

- FCFS: first come first served
- RR: RoundRobin
- SPN: Shortest Process Next
- SRT: Shortest Remaining Time
- HRRN: Highest Response Ratio Next
- FB: Feedback

	FCFS	Round Robin	SPN	SRT	HRRN	Feedback
Selection Function	max[w]	constant	min[s]	min[s − e]	$\max\left(\frac{w+s}{s}\right)$	(see text)
Decision Mode	Non-preemptive	Preemptive (at time quantum)	Non-preemptive	Preemptive (at arrival)	Non-preemptive	Preemptive (at time quantum)
Throughput	Not emphasized	May be low if quantum is too small	High	High	High	Not emphasized
Response Time	May be high, especially if there is a large variance in process execution times	Provides good response time for short processes	Provides good response time for short processes	Provides good response time	Provides good response time	Not emphasized
Overhead	Minimum	Minimum	Can be high	Can be high	Can be high	Can be high
Effect on Processes	Penalizes short processes; penalizes I/O-bound processes	Fair treatment	Penalizes long processes	Penalizes long processes	Good balance	May favor I/O-bound processes
Starvation	No	No	Possible	Possible	No	Possible

Selection Function

选择 函数

- **Determines which process, among ready processes, is selected next for execution**

- May be based on priority, resource requirements, or the execution characteristics of the process

- If based on **execution characteristics** then important quantities are:
  - w = time spent in system so far, waiting 已等待的时间
  - e = time spent in execution so far 已执行的时间
  - s = total service time required by the process, including e; generally, this quantity must be estimated or supplied by the user

全部服务时间

Decision Mode（决策模式）

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- **The decision mode specifies the instants in time at which the selection function is exercised.**

- **Two categories:**
  - **Nonpreemptive**
    - ➤ once a process is in the running state, it will continue until it terminates or blocks itself for I/O
  - **Preemptive**:
    - ➤ currently running process may be interrupted and moved to ready state by the OS
    - ➤ preemption may occur when new process arrives, on an interrupt, or periodically

Nonpreemptive vs Preemptive

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- **Nonpreemptive**
  - once a process is in the running state, it will continue until it terminates or blocks itself for I/O
- **Preemptive(抢占)**
  - currently running process may be interrupted and moved to ready state by the OS
  - preemption may occur when new process arrives, on an interrupt, or periodically

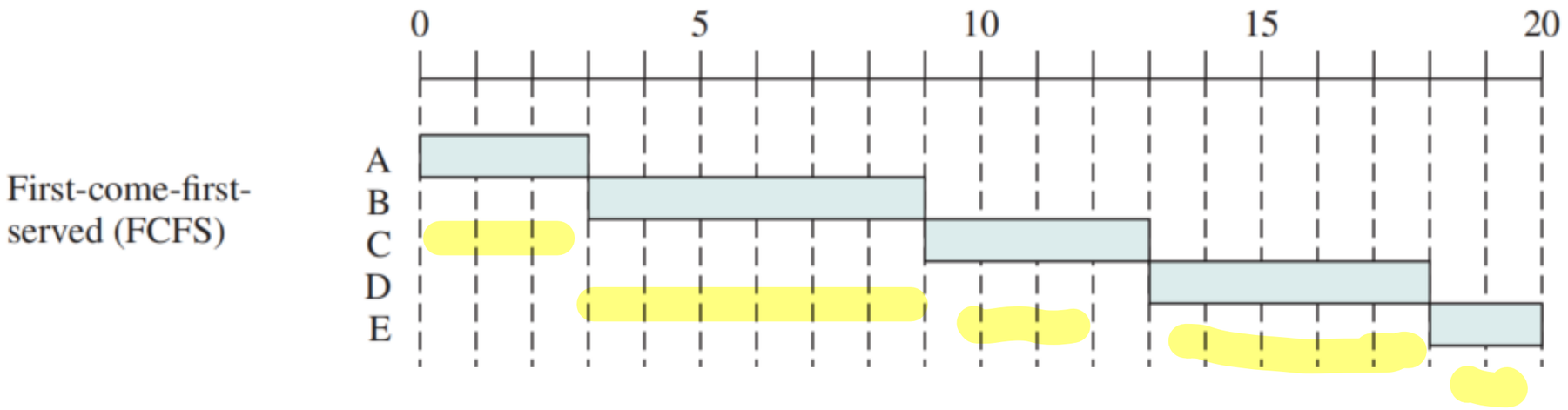
新进程到达 打断 周期性触发

Process Scheduling Example

Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

First-Come-First-Served (FCFS)

- Simplest scheduling policy
- Also known as first-in-first-out (FIFO) or a strict queuing scheme
- When the current process ceases to execute, the longest process in the Ready queue is selected



Issues of FCFS

- FCFS performs much better for long processes than short ones.

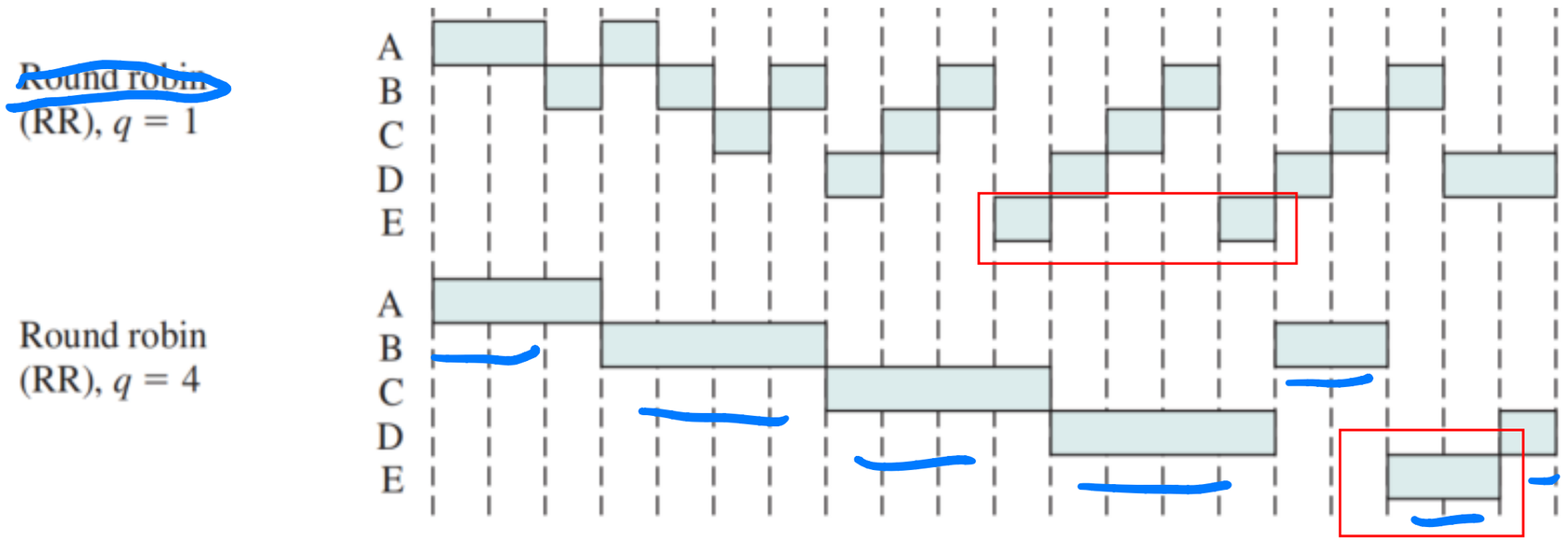
Process	Arrival Time	Service Time (T_s)	Start Time	Finish Time	Turnaround Time (T_r)	T_r/T_s
W	0	1	0	1	1	1
X	1	100	1	101	100	1
Y	2	1	101	102	100	100
Z	3	100	102	202	199	1.99
Mean					100	26

- Tends to favor processor-bound processes over I/O-bound processes

优待处理器型进程，而非I/O设备。

Round Robin 时间片轮转

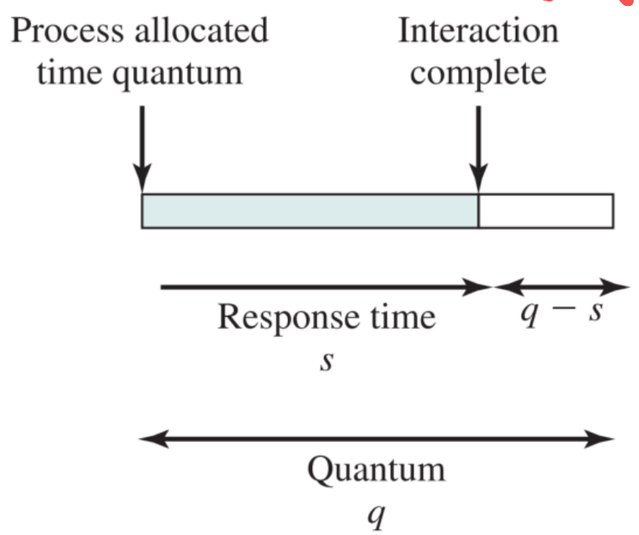
- Uses preemption based on a clock
- Also known as time slicing because each process is given a slice of time before being preempted
- Particularly effective in a general-purpose time-sharing system or transaction processing system
- Principal design issue is the length of the time quantum, or slice, to be used



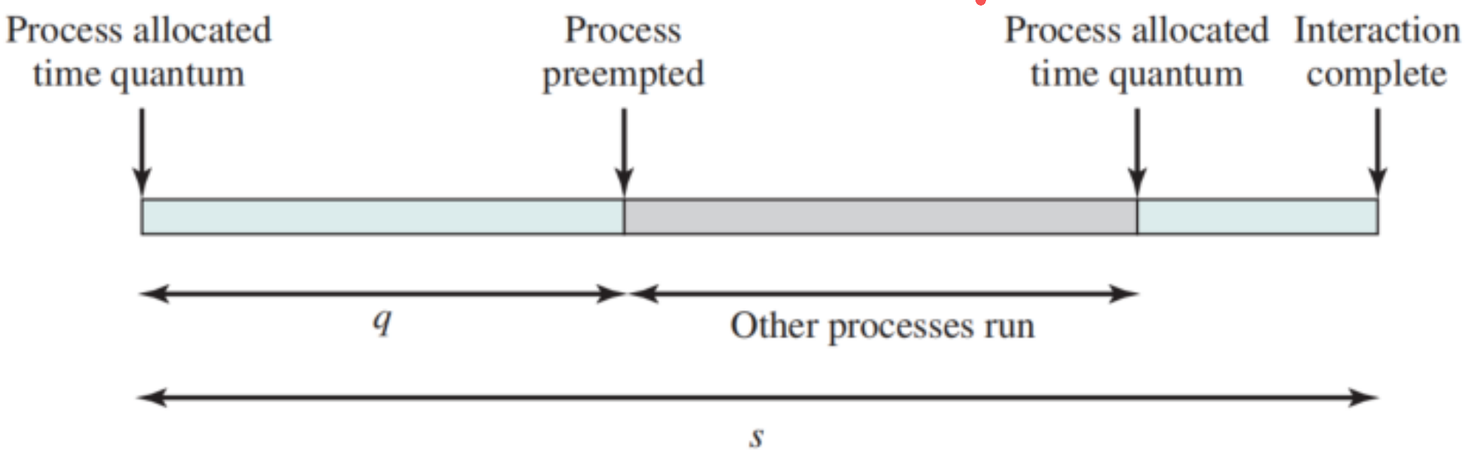
- Short process are processed quickly
- Switch overhead

Effect of Size of Preemption Time Quantum

- Very short time quanta should be avoided
- One useful guide is that the time quantum should be slightly greater than the time required for a typical interaction or process function



(a) Time quantum greater than typical interaction



(b) Time quantum less than typical interaction

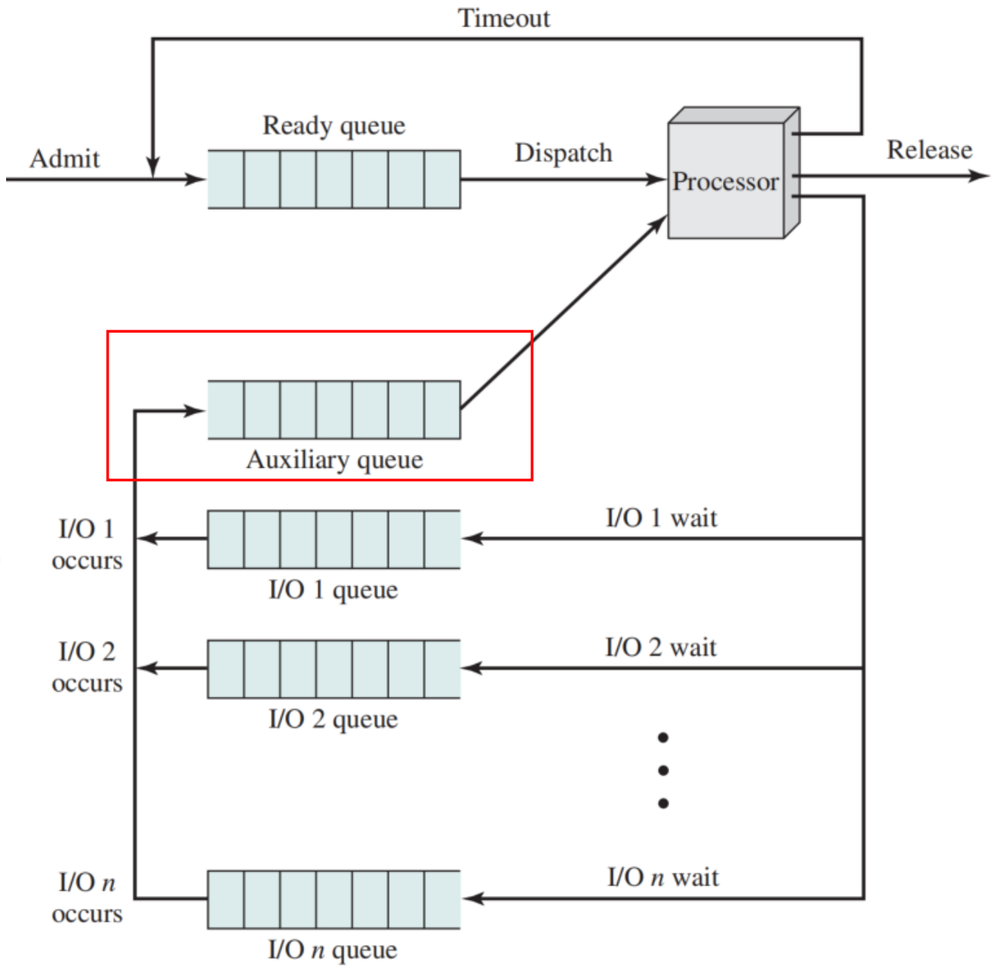
Virtual Round-Robin Scheduler

虚拟时间片轮换

- **One drawback is its relative treatment of processor-bound and I/O-bound processes**
  - An I/O-bound process uses a processor for a short period, then is blocked for I/O; it waits for the I/O operation to complete then joins the ready queue.
  - a processor-bound process generally uses a complete time quantum while executing and immediately returns to the ready queue.

缺点

1.Processes in the auxiliary queue get **preference over** those in the main ready queue.
2.Run time = q-e

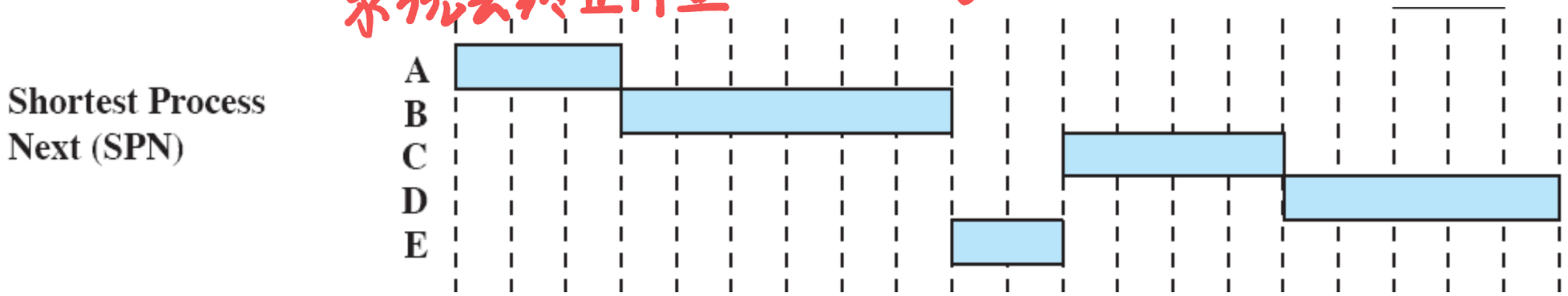


Queueing Diagram for Virtual Round-Robin (VRR) Scheduler

VRR改进策略：辅助队列中的进程优先于主就绪队列

Shortest Process Next (SPN)

- **Nonpreemptive policy in which the process with the shortest expected processing time is selected next**
  - A short process will jump to the head of the queue
  - Possibility of starvation for longer processes
  - One difficulty is the need to know, or at least estimate, the required processing time of each process
  - If the programmer's estimate is substantially under the actual running time, the system may abort the job



Estimate running time

❑Simple averaging

$$S_{n+1} = \frac{1}{n}\sum T_i \quad \Rightarrow \quad S_{n+1} = \frac{1}{n}T_n + \frac{n-1}{n}S_n$$

equal weight

❑Exponential averaging, 指数平均 $\alpha \in (0,1)$

- give greater weight to more recent instances, because these are more likely to reflect future behavior.

反映未来

$$S_{n+1} = \alpha T_n + (1-\alpha)S_n$$

$$S_{n+1} = \alpha T_n + (1-\alpha)\alpha T_{n-1} \ldots + (1-\alpha)^n S_1$$

$\alpha = 0.8$

$$S_{n+1} = 0.8T_n + 0.16T_{n-1} + 0.032T_{n-2} + 0.0064T_{n-3} + \ldots + (0.2)^n S_1$$

Exponential Smoothing Coefficients

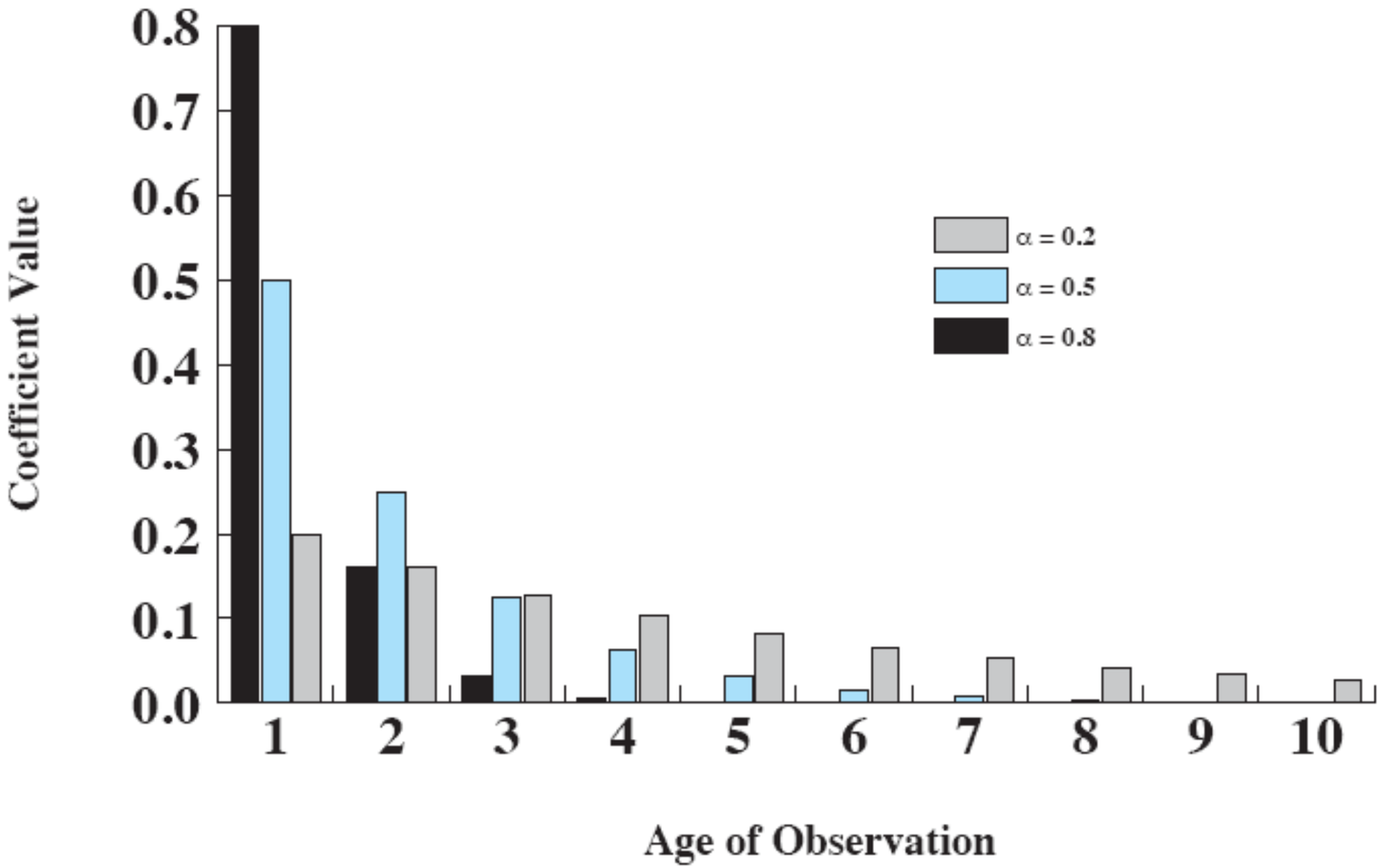
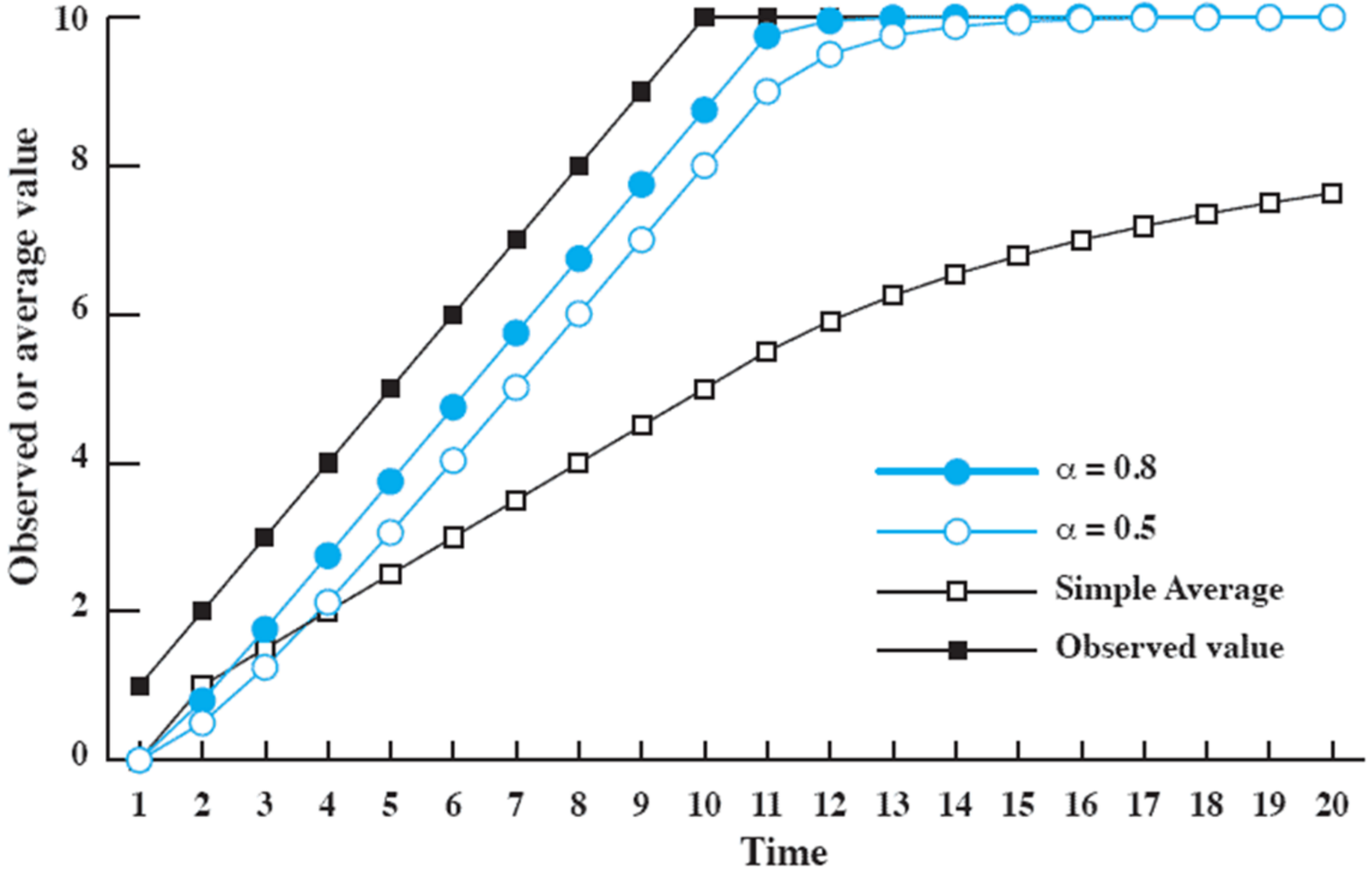


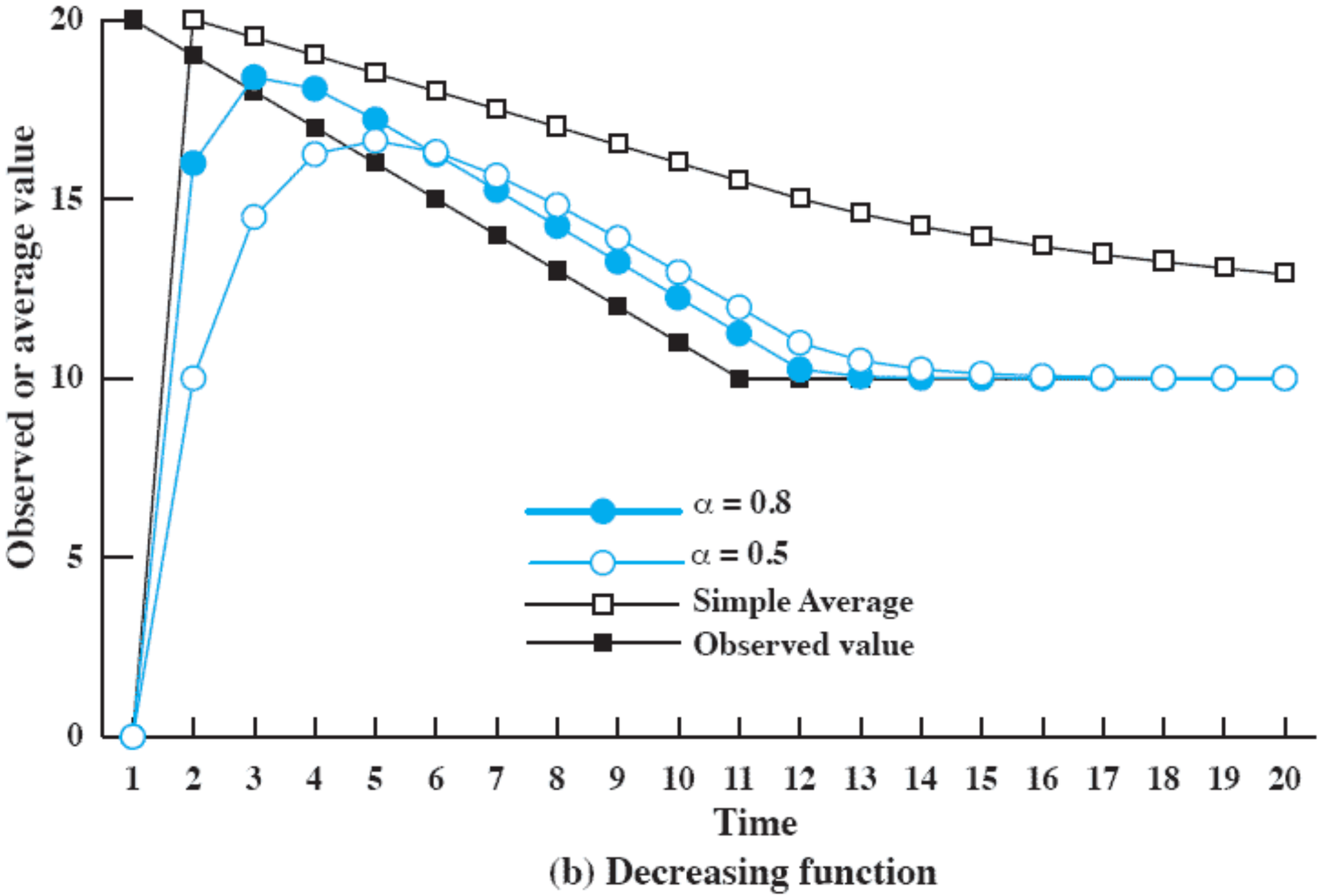
Figure 9.8 Exponential Smoothing Coefficients

Use Of Exponential Averaging



(a) Increasing function

Use Of Exponential Averaging



(b) Decreasing function

Shortest Remaining Time (SRT)

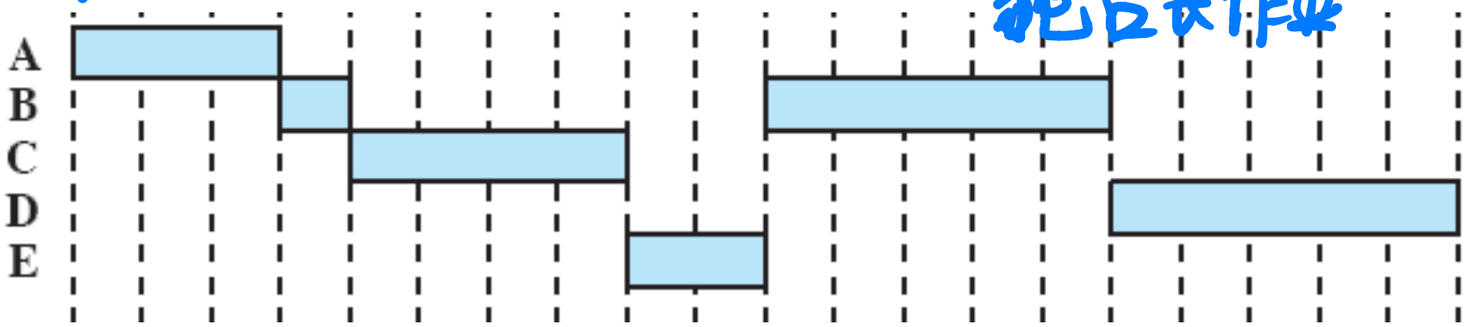
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最短剩余时间优先

- **Preemptive version of SPN**
- **Chooses the process that has the shortest expected remaining processing time**
  - have to estimate the processing time
  - service times must be recorded, contributing to overhead
  - Risk of starvation of longer processes
  - Should give superior turnaround time performance to SPN because a short job is given immediate preference to a running longer job

周转时间性能优于 SPN，因为短作业可以立即抢占长作业

Shortest Remaining Time (SRT)



Highest Response Ratio Next (HRRN)

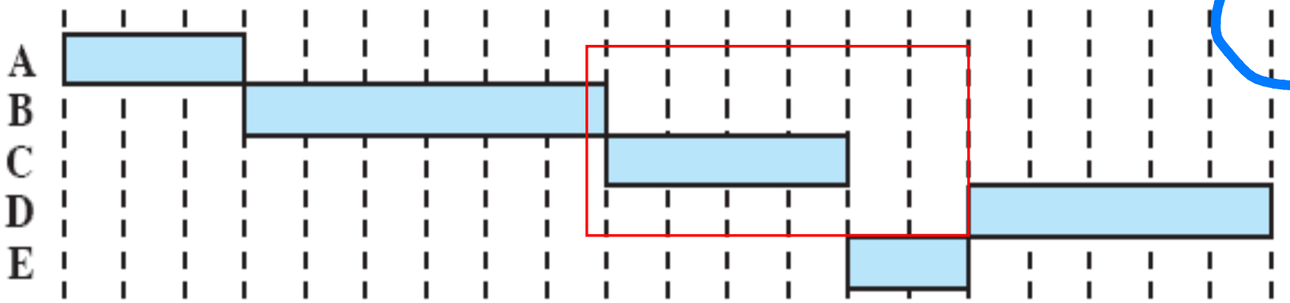
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- **Chooses next process with the greatest ratio:** $\max R=\frac{w+s}{s}$, $R^{min}=1$
  - Accounts for the age of the process
  - Need to estimated the expected service time

$$Ratio = \frac{time\ spent\ waiting + expected\ service\ time}{expected\ service\ time}$$

- **While shorter jobs are favored, aging without service increases the ratio so that a longer process will eventually get past competing shorter jobs**

Highest Response Ratio Next (HRRN)



Non-preemptive

Feedback Scheduling

- If we have no indication of the relative length of various processes, then none of SPN, SRT, and HRRN can be used
- FB: When a process first enters the system, it is placed in RQ0. After its first preemption, when it returns to the Ready state, it is placed in RQ1.
  - a preference for shorter jobs
  - focus on the time spent in execution so far

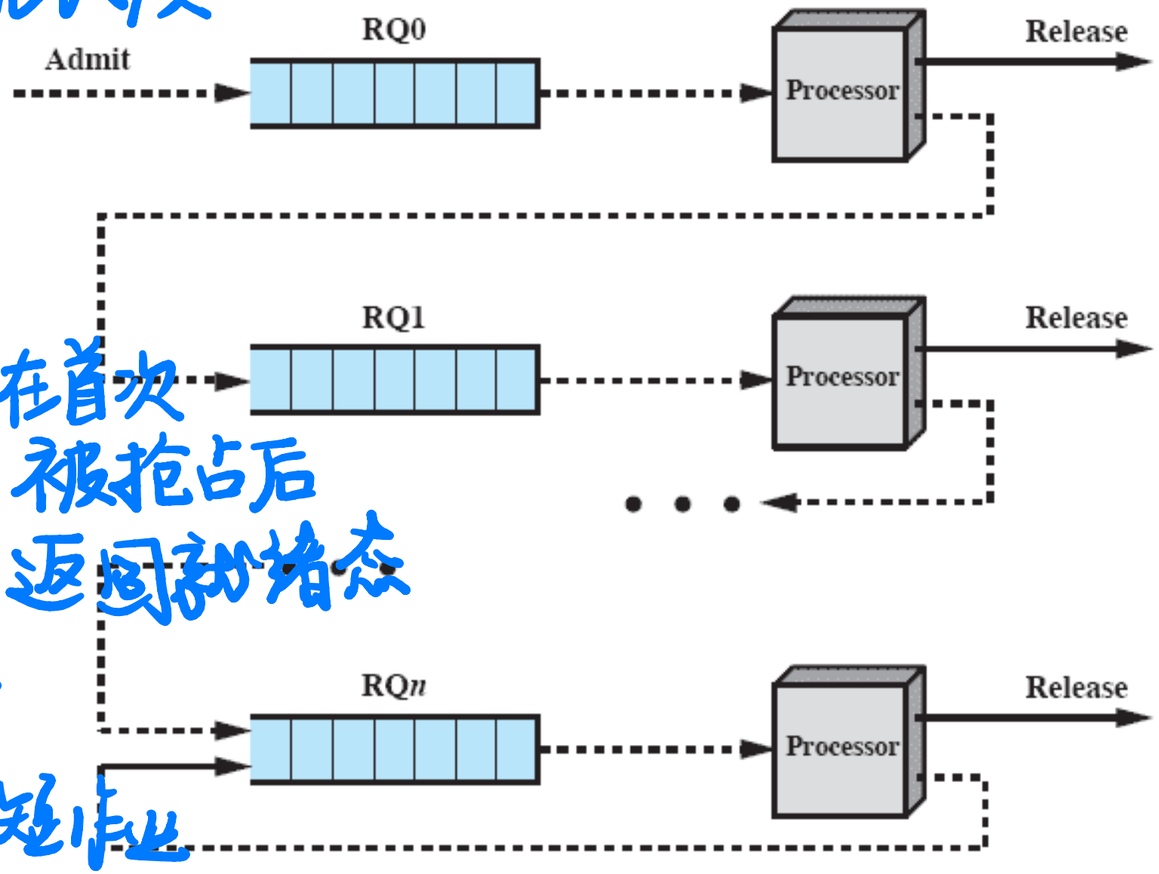


Figure 9.10 Feedback Scheduling

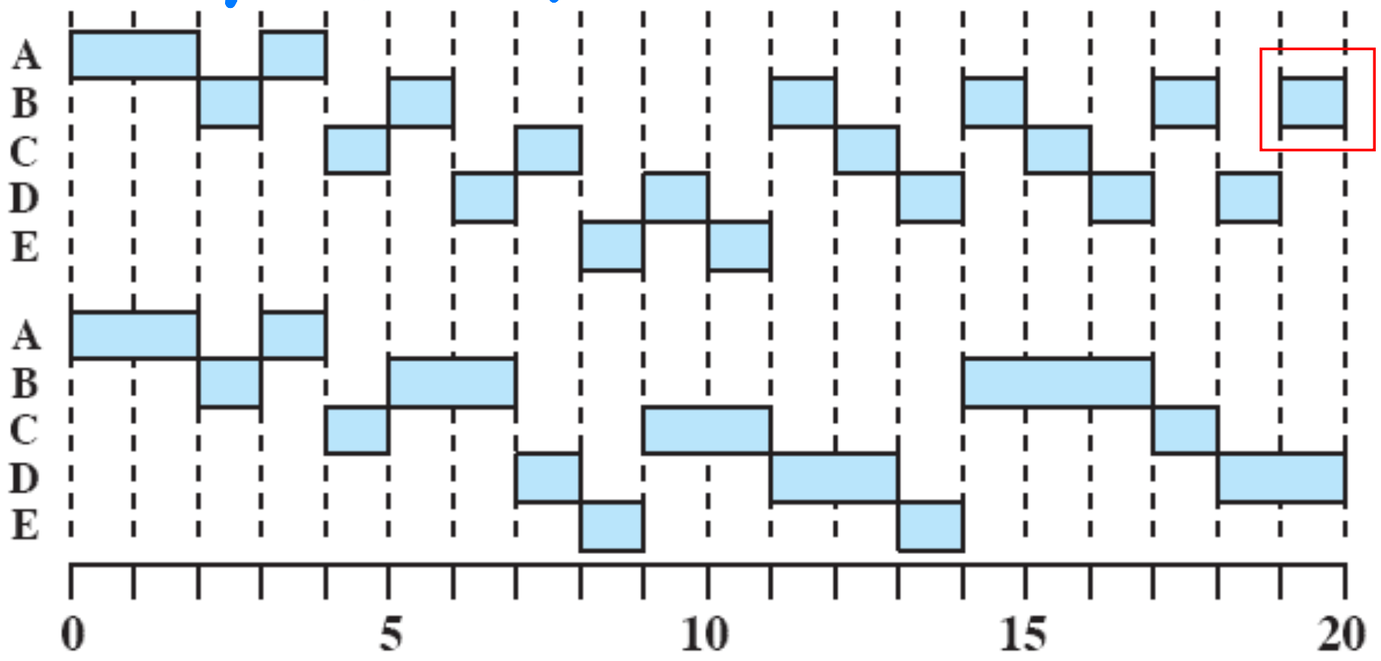
Feedback Performance

Issues: Turnaround time of longer processes is too long in q=1

时间片q=1时，长进程周转时间过长

Feedback
$q = 1$

Feedback
$q = 2^i$



In general, a process scheduled from RQ_i is allowed to execute $q = 2^i$ time units before preemption

被允许执行$q=2^i$个时间单位后被抢占

A Comparison of Scheduling Policies

Process	A	B	C	D	E	
Arrival Time	0	2	4	6	8	
Service Time (T_s)	3	6	4	5	2	Mean
FCFS						
Finish Time	3	9	13	18	20	
Turnaround Time (T_r)	3	7	9	12	12	8.60
T_r/T_s	1.00	1.17	2.25	2.40	6.00	2.56
RR q = 1						
Finish Time	4	18	17	20	15	
Turnaround Time (T_r)	4	16	13	14	7	10.80
T_r/T_s	1.33	2.67	3.25	2.80	3.50	2.71
RR q = 4						
Finish Time	3	17	11	20	19	
Turnaround Time (T_r)	3	15	7	14	11	10.00
T_r/T_s	1.00	2.5	1.75	2.80	5.50	2.71
SPN						

A Comparison of Scheduling Policies （cont.）

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SPN						
Finish Time	3	9	15	20	11	
Turnaround Time (T_r)	3	7	11	14	3	7.60
T_r/T_s	1.00	1.17	2.75	2.80	1.50	1.84
SRT						
Finish Time	3	15	8	20	10	
Turnaround Time (T_r)	3	13	4	14	2	7.20
T_r/T_s	1.00	2.17	1.00	2.80	1.00	1.59
HRRN						
Finish Time	3	9	13	20	15	
Turnaround Time (T_r)	3	7	9	14	7	8.00
T_r/T_s	1.00	1.17	2.25	2.80	3.5	2.14
FB $q = 1$						
Finish Time	4	20	16	19	11	
Turnaround Time (T_r)	4	18	12	13	3	10.00
T_r/T_s	1.33	3.00	3.00	2.60	1.5	2.29
FB $q = 2^i$						
Finish Time	4	17	18	20	14	
Turnaround Time (T_r)	4	15	14	14	6	10.60
T_r/T_s	1.33	2.50	3.50	2.80	3.00	2.63

Fair-Share Scheduling

公平 共享 调度

- **Scheduling decisions based on the process sets 进程组**
  - user applications or jobs may be organized as multiple processes
- **Each user is assigned a share (份额) of the processor**
  - e.g. four groups, each with 25% of the processor usage
- **Objective is to monitor usage to give fewer resources to users who have had more than their fair share and more to those who have had less than their fair share**

Fair-Share Scheduler Implemented in UNIX

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- fair-share scheduler (FSS): divides the user community into a set of fair-share groups and allocates a fraction of the processor resource to each group.

分配一定比例的处理器资源

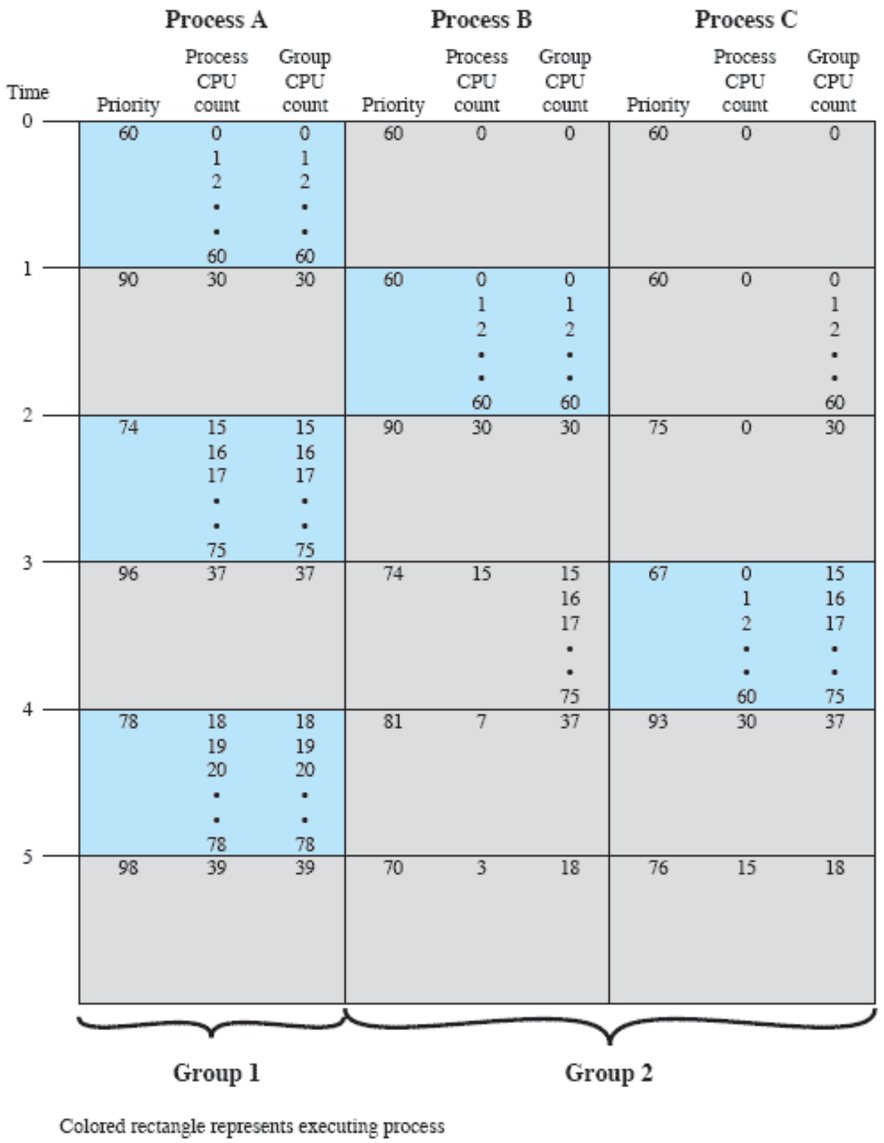
- Scheduling is done on the basis of priority, the higher the numerical value of the priority, the lower is the priority.

优先级数值级别小

- Account for the priority of the process, its processor usage, and the processor usage of the group to which the process belongs,

Fair-Share Scheduler (FSS)

西南交通大學 Southwest Jiaotong University



Time	Process A Priority	Process A Process CPU count	Process A Group CPU count	Process B Priority	Process B Process CPU count	Process B Group CPU count	Process C Priority	Process C Process CPU count	Process C Group CPU count
0	60	0 1 2 • • 60	0 1 2 • • 60	60	0	0	60	0	0
1	90	30	30	60	0 1 2 • • 60	0 1 2 • • 60	60	0	0 1 2 • • 60
2	74	15 16 17 • • 75	15 16 17 • • 75	90	30	30	75	0	30
3	96	37	37	74	15	15 16 17 • • 75	67	0 1 2 • • 60	15 16 17 • • 75
4	78	18 19 20 • • 78	18 19 20 • • 78	81	7	37	93	30	37
5	98	39	39	70	3	18	76	15	18

Group 1 Group 2

Colored rectangle represents executing process

Figure 9.16 Example of Fair Share Scheduler—Three Processes, Two Groups

Summary

- **The operating system must make three types of scheduling decisions with respect to the execution of processes:**
  - Long-term – determines when new processes are admitted to the system
  - Medium-term – part of the swapping function and determines when a program is brought into main memory so that it may be executed
  - Short-term – determines which ready process will be executed next by the processor
- **From a user's point of view, response time is generally the most important characteristic of a system; from a system point of view, throughput or processor utilization is important**
- **Scheduling Policies:**
  - FCFS, Round Robin, SPN, SRT, HRRN, Feedback

Quiz

9.2. What factors determine the time quantum in round robin scheduling? Consider the system:

Process	Arrival Time	Processing Time
P1	0	12
P2	2	6
P3	8	18
P4	10	4

上下切换

Context switch takes a time of 1 unit. Compute the average turnaround time of the processes for the time quanta $q = 2, q = 4$, and $q = 8$ respectively.

决定时间量子的关键因素：
① CPU效率与系统响应时间：
时间量子过小：过多的上下文切换（CPU时间被浪费）
大：降低系统响应速度
② 典型进程用CPU突发时间
③ 系统负载
④、上下文切换

Quiz

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